Limits – 3

Asymptotics of sequences

Homework

1 сентября 2026 года

Theoretical minimum. The lemma on a neutral fixed point $x_{n+1} = x_n - ax_n^p + o(x_n^p)$; the Stolz–Cesàro theorem; telescoping in powers of x_n; comparison of a sum with an integral; the asymptotics $H_n = \ln n + \gamma + O(1/n)$; the step from "the limit is zero" to the leading term together with an estimate of the remainder.

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1. Let $x_0 \in [-1, 1]$ and $x_{n+1} = x_n - \arcsin(\sin^3 x_n)$ for $n \geqslant 0$. Compute $\lim_{n\to\infty} \sqrt{n}\, x_n$.

2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and suppose the sequence $a_n = \int_0^1 f(n + x)\, dx$ converges. Prove that the sequence $b_n = \int_0^1 f(nx)\, dx$ converges as well.

3. (VJIMC 1999) Find the limit $\lim\limits_{n\to\infty} \Big(\prod\limits_{k=1}^{n} \dfrac{k}{k+n}\Big)^{e^{1999/n}-1}$.

4. (IMC 2025 shortlist) Compute $\lim\limits_{n\to\infty} \dfrac{n^1 + n^2 + \cdots + n^n}{1^n + 2^n + \cdots + n^n}$.

5. (SEEMOUS) Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous strictly decreasing function with $f([0, 1]) \subseteq [0, 1]$.

   a) Prove that for every $n \geqslant 1$ the equation $f(x) = x^n$ has a unique solution $a_n \in (0, 1)$, and that $\lim_{n\to\infty} a_n = 1$.

   b) Suppose in addition that f has a continuous derivative, $f(1) = 0$ and $f'(1) < 0$, and put $F(x) = \int_x^1 f(t)\, dt$ for $x \in \mathbb{R}$. Study the convergence of the series $\sum_{n\geqslant 1} \big(F(a_n)\big)^{\alpha}$, with $\alpha \in \mathbb{R}$.

6. (IMC 2024) Define $x_1 = 2$, $x_2 = 4$ and $x_{n+2} = 3x_{n+1} - 2x_n + \frac{2^n}{x_n}$ for $n \geqslant 1$. Prove that $\lim_{n\to\infty} \frac{x_n}{2^n}$ exists and satisfies

$$\frac{1+\sqrt{3}}{2} \leqslant \lim_{n\to\infty} \frac{x_n}{2^n} \leqslant \frac{3}{2}.$$

7. (OMOUS 2024) For an integer $n \geqslant 2$ let $I(n) = \displaystyle\int_0^{\pi/2} \frac{\{nx\}}{\sin x}\,dx$, where $\{s\}$ is the fractional part of s.

   a) Prove that the integral $I(n)$ is convergent.

   b) Prove that there exist constants $C_1 > 0$ and $C_2 > 0$ such that $C_1 \ln n \leqslant I(n) \leqslant C_2 \ln n$ for all $n \geqslant 2$.

8. (Vienna MC 2025) Let (a_n) satisfy $a_0 > 0$ and $a_{n+1} = \frac{a_n}{a_n^2 + a_n + 1}$ for all n. Show that $a_n \to 0$ and determine $\lim_{n\to\infty} n a_n$.

9. (SEEMOUS) For $n \geqslant 1$ put

$$x_n = \int_0^1 \ln\bigl(1 + x + x^2 + \cdots + x^n\bigr) \cdot \ln \frac{1}{1-x}\,dx.$$

   a) Show that x_n is finite for every n and that $\lim_{n\to\infty} x_n = 2$.

   b) Compute $\lim_{n\to\infty} \frac{n}{\ln n}\,(2 - x_n)$.

10. (VJIMC 2002) Prove that $\displaystyle\lim_{n\to\infty} n^2\Bigl(\int_0^1 \sqrt[n]{1+x^n}\,dx - 1\Bigr) = \frac{\pi^2}{12}$.